

$T = 1.5 \leq 4 \Rightarrow \text{Reject } H_0 \Rightarrow \text{We have sufficient evidence to say that two population distributions are different.}$

DATA SET

DS1511

17. Machine Breakdowns The number of machine breakdowns per month was recorded for 9 months on two identical machines, A and B, used to make wire rope:

Month	A	B
1	3	7
2	14	12
3	7	9
4	10	15
5	9	12
6	6	6
7	13	12
8	6	5
9	7	13

- Do the data provide sufficient evidence to indicate a difference in the monthly breakdown rates for the two machines? Test by using a value of α near .05.
- Can you think of a reason the breakdown rates for the two machines might vary from month to month?

b. Workload on each machine may vary from month to month.

$$(A). H_0: p = \frac{1}{2} \quad H_a: p \neq \frac{1}{2}$$

A-B	-4	2	-2	-5	-3	1	1	-6
rank	6	3.5	3.5	7	5	1.5	1.5	8

$$T^+ = 3.5 + 1.5 + 1.5 = 6.5$$

$$T^- = 6 + 3.5 + 7 + 5 + 8 = 29.5$$

$$T = \min(T^+, T^-) = 6.5$$

$$\alpha = 0.05, \quad n = 8$$

$$RR: \{T \mid T \leq 4\}$$

$$6.5 > 4 \Rightarrow \text{non-reject } H_0$$

There is no sufficient evidence to say the monthly breakdown rate for the two machines are different.